Exp No:12

Determination and plotting of magnetic and electric field intensity due to infinite long conductor at any specific point located at H

Date:

Aim:

To write program for Magnetic field intensity of infinite long conductor using SCILAB.

Software required:

- SCILAB version 6.0.1

Theory:

In electromagnetism, the fields around a conductor carrying a current are fundamental concepts. For an infinite long conductor carrying a current, we can determine both the **magnetic field intensity** and **electric field intensity** at a point located at a distance from the conductor.

Magnetic Field Intensity due to an Infinite Long Conductor

Ampère's Law:

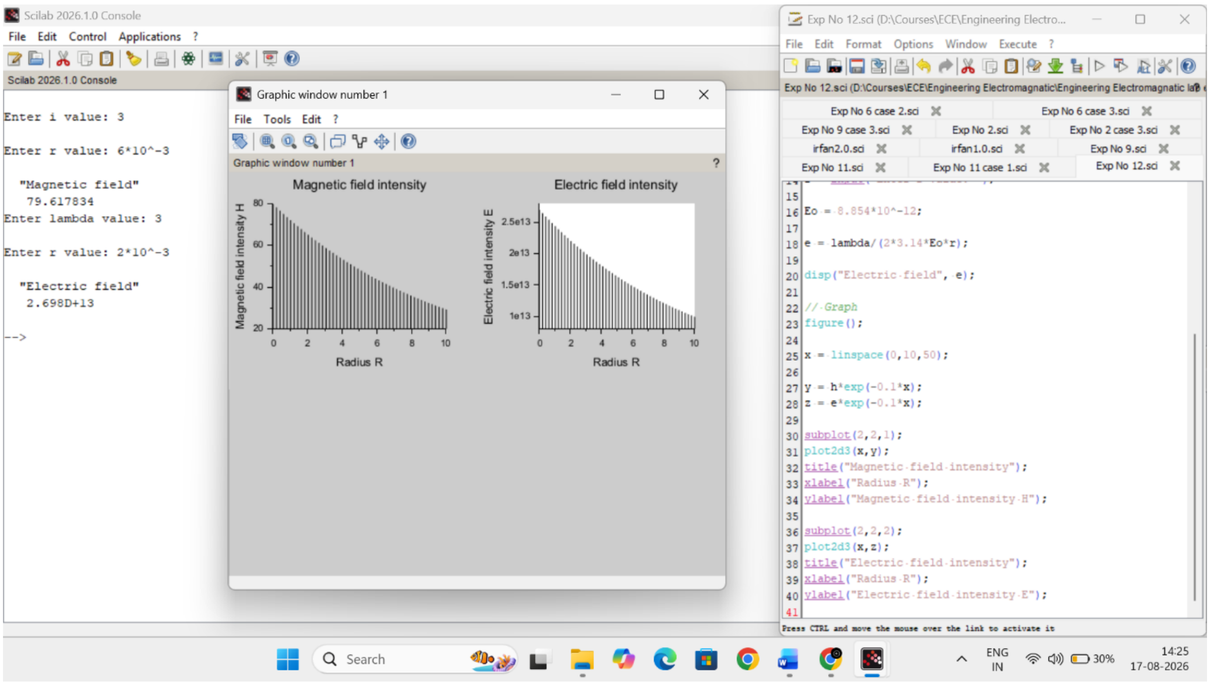
For a steady current, the magnetic field around a conductor can be determined using **Ampère's Law**.

The law states:

$$\oint \vec{H} \cdot d\vec{l} = I_{\text{enc}}$$

Where:

- $\vec{H}$ is the magnetic field intensity,
- $d\vec{l}$ is the differential length along the path of integration (typically a circle around the wire),
- I_{enc} is the current enclosed by the path (in this case, the current I in the wire).


Scilab 2026.1.0 Console
File Edit Control Applications ?
Enter i value: 3
Enter r value: 6*10^-3
"Magnetic field"
79.617834
Enter lambda value: 3
Enter r value: 2*10^-3
"Electric field"
2.698D+13
-->
Graphic window number 1
File Tools Edit ?
Magnetic field intensity
Magnetic field intensity H
Radius R
Electric field intensity
Electric field intensity E
Radius R
Exp No 12.sci (D:\Courses\ECE\Engineering Electro...
File Edit Format Options Window Execute ?
Exp No 6 case 2.sci
Exp No 6 case 3.sci
Exp No 9 case 3.sci
Exp No 2.sci
Exp No 2 case 3.sci
irfan2.0.sci
irfan1.0.sci
Exp No 9.sci
Exp No 11.sci
Exp No 11 case 1.sci
Exp No 12.sci
Eo = 8.854*10^-12;
e = lambda/(2*3.14*Eo*r);
disp("Electric field", e);
// Graph
figure();
x = linspace(0,10,50);
y = h*exp(-0.1*x);
z = e*exp(-0.1*x);
subplot(2,2,1);
plot2d3(x,y);
title("Magnetic field intensity");
xlabel("Radius R");
ylabel("Magnetic field intensity H");
subplot(2,2,2);
plot2d3(x,z);
title("Electric field intensity");
xlabel("Radius R");
ylabel("Electric field intensity E");
Press CTRL and move the mouse over the link to activate it
Search
ENG
IN
30%
14:25
17-08-2026

EXP NO:- 12 Determine magnetic and electric field intensity.

$i = 3\ A$ $\lambda = 3\ m$

$r = 6\ mm$

$z = 2\ mm$

magnetic field Intensity

$$H = \frac{I}{2\pi r}$$

$$H = \frac{3}{2\pi(6\times10^{-3})}$$

$$H = \frac{3}{0.037699}$$

$$\boxed{H = 79.617\ A/m}$$

Electric field Intensity

$$E = \frac{\lambda}{2\pi \varepsilon_0 r} \qquad \varepsilon_0 = 8.854\times10^{-12}\ F/$$

$$E = \frac{3}{2\pi(8.854\times10^{-12})(2\times10^{-3})}$$

$2\pi\ (8.854\times10^{-12})(2\times10^{-3}) = 1.1127\times10^{-13}$

$$E = \frac{3}{1.1127\times10^{-13}}$$

$$\boxed{E = 2.698\times10^{13}}$$

O/P

Result:

Thus the program was verified for Magnetic field intensities in ferromagnetic materials using SCILAB was successfully.

Case 1: To examine how the magnetic and electric field intensities change with varying distances from an infinitely long current-carrying conductor.

Analytical:

Calculate the magnetic and electric field intensities change with varying distances from an infinitely long current-carrying conductor for following conditions

- **Conductor**: Infinite long conductor carrying a steady current III.
- **Distance (H)**: The distance from the conductor varies from 0.1 m to 10 m (i.e., 0.1 m, 1 m, 5 m, 10 m).
- **Current (I)**: Constant current of 10 A.

Solution:

$$H = \frac{I}{2\pi r}$$

1. For $r = 0.1\,\mathrm{m}$:

$$H = \frac{10}{2\pi(0.1)} = \frac{10}{0.628} \approx 15.92\,\mathrm{A/m}$$

2. For $r = 1\,\mathrm{m}$:

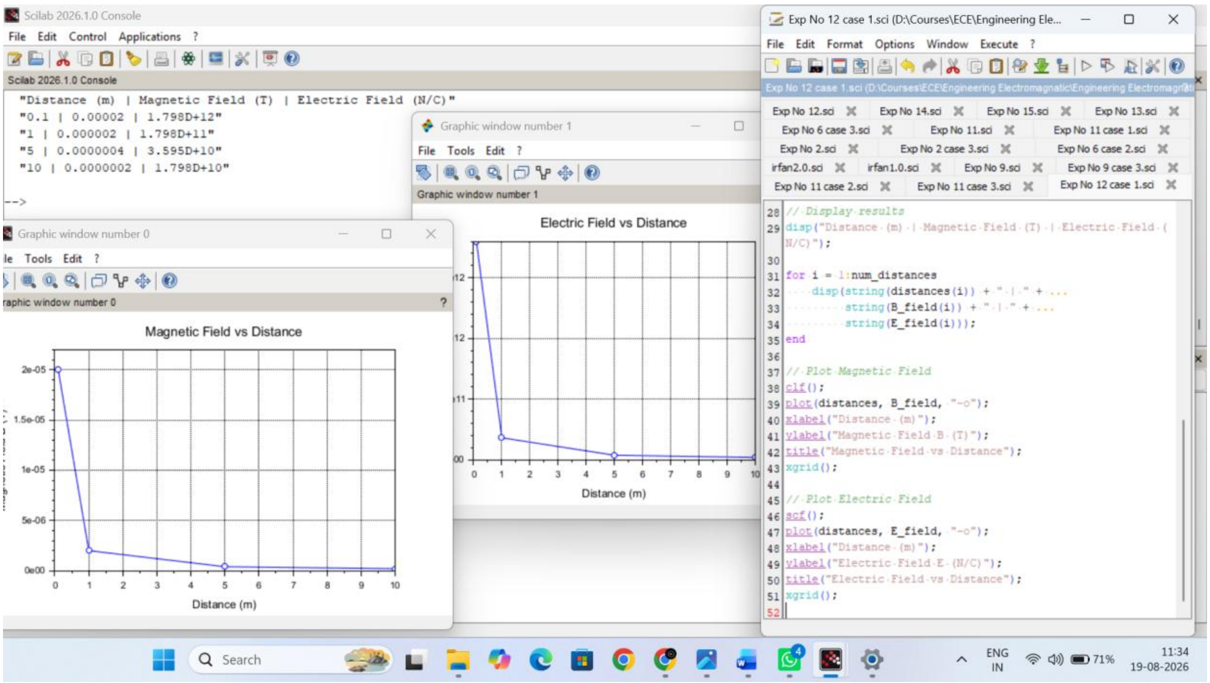
$$H = \frac{10}{2\pi(1)} = \frac{10}{6.283} \approx 1.59\,\mathrm{A/m}$$

3. For $r = 5\,\mathrm{m}$:

$$H = \frac{10}{2\pi(5)} = \frac{10}{31.415} \approx 0.318\,\mathrm{A/m}$$

4. For $r = 10\,\mathrm{m}$:

$$H = \frac{10}{2\pi(10)} = \frac{10}{62.832} \approx 0.159\,\mathrm{A/m}$$



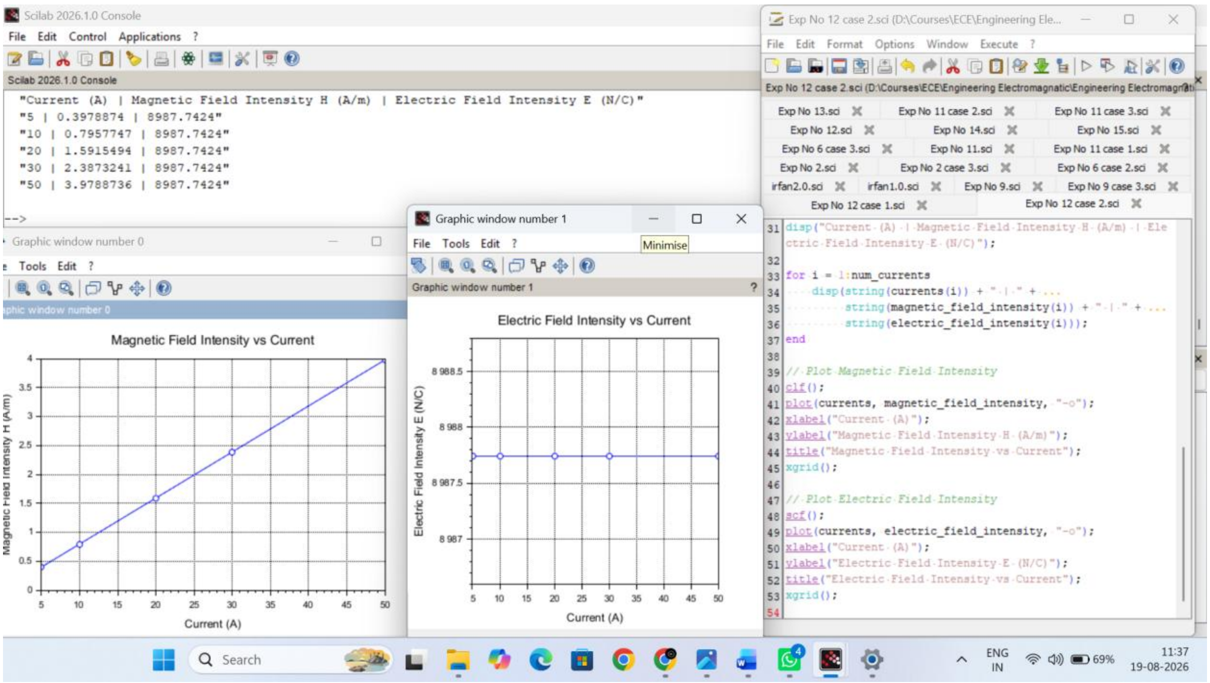
Case 2: To study how the magnetic and electric field intensities change for an infinitely long conductor at a fixed distance H but with varying currents.

Analytical:

To calculate magnetic and electric field intensities change for an infinitely long conductor at a fixed distance H but with varying currents for the following,

- Conductor: Infinite long conductor.
- Distance (H): Fixed at 2 meters.
- Current (I): Varying from 5 A to 50 A (i.e., 5 A, 10 A, 20 A, 30 A, 50 A).

Solution:



Case 3: To analyze the magnetic and electric field intensities due to an infinitely long conductor when there is a potential difference (voltage) across the conductor, simulating a capacitive effect.

Analytical:

Calculate the magnetic and electric field intensities due to an infinitely long conductor when there is a potential difference (voltage) across the conductor, simulating a capacitive effect.

- **Conductor**: Infinite long conductor.
- **Voltage (V)**: Varies from 100 V to 1000 V (i.e., 100 V, 300 V, 500 V, 700 V, 1000 V).
- **Distance (H)**: Fixed at 1 m.
- **Current (I)**: Varies depending on the voltage (using Ohm's Law, I=V/R, where resistance R is assumed).

Solution:

1. For $V = 100\ \mathrm{V}$:

- Current $I = 100\ \mathrm{A}$
- Magnetic field intensity $H = \frac{I}{2\pi r} = \frac{100}{2\pi(1)} = \frac{100}{6.283} \approx 15.92\ \mathrm{A/m}$
- Electric field intensity $E = \frac{V}{r} = \frac{100}{1} = 100\ \mathrm{V/m}$

2. For $V = 300\ \mathrm{V}$:

- Current $I = 300\ \mathrm{A}$
- Magnetic field intensity $H = \frac{I}{2\pi r} = \frac{300}{2\pi(1)} = \frac{300}{6.283} \approx 47.75\ \mathrm{A/m}$
- Electric field intensity $E = \frac{V}{r} = \frac{300}{1} = 300\ \mathrm{V/m}$

3. For $V = 500\ \mathrm{V}$:

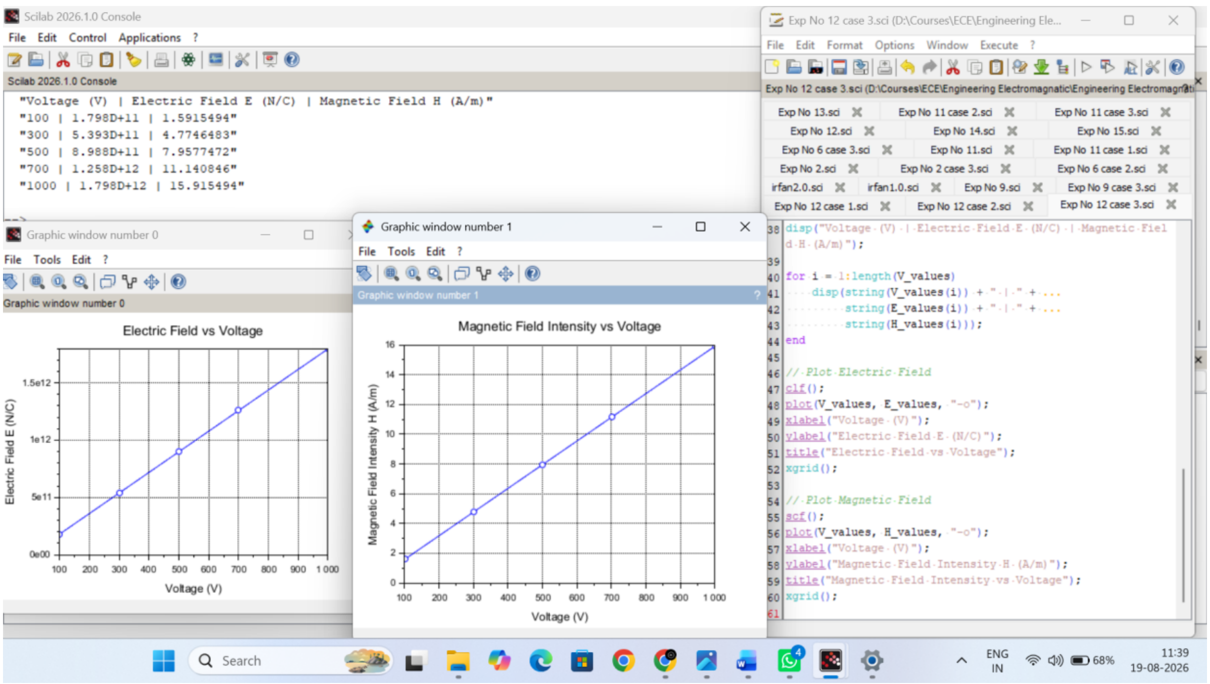
- Current $I = 500\ \mathrm{A}$
- Magnetic field intensity $H = \frac{I}{2\pi r} = \frac{500}{2\pi(1)} = \frac{500}{6.283} \approx 79.58\ \mathrm{A/m}$
- Electric field intensity $E = \frac{V}{r} = \frac{500}{1} = 500\ \mathrm{V/m}$

4. For $V = 700\ \mathrm{V}$:

- Current $I = 700\ \mathrm{A}$
- Magnetic field intensity $H = \frac{I}{2\pi r} = \frac{700}{2\pi(1)} = \frac{700}{6.283} \approx 111.46\ \mathrm{A/m}$
- Electric field intensity $E = \frac{V}{r} = \frac{700}{1} = 700\ \mathrm{V/m}$

5. For $V = 1000\ \mathrm{V}$:

- Current $I = 1000\ \mathrm{A}$
- Magnetic field intensity $H = \frac{I}{2\pi r} = \frac{1000}{2\pi(1)} = \frac{1000}{6.283} \approx 159.15\ \mathrm{A/m}$
- Electric field intensity $E = \frac{V}{r} = \frac{1000}{1} = 1000\ \mathrm{V/m}$



Result:

Thus the program was verified for the magnetic and electric field intensities due to infinitely long conductor.